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# COMMON QUARTERLY EXAMINATION - 2025

## Standard XII

Reg.No. | | | | | | |

## MATHEMATICS

*Time : 3.00 hrs* *Marks : 90*

**Part - I** **20 x 1 = 20**

**I. Answer all the questions.**

1. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then $\lambda$ is

   a) 19 b) 17 c) 14 d) 21

2. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies

   a) $|k| > 6$ b) $|k| \geq 6$ c) $k = 0$ d) $|k| \leq 6$

3. The conjugate of a complex number is $\frac{1}{i-2}$. Then, the complex number is

   a) $\frac{-1}{i+2}$ b) $\frac{1}{i-2}$ c) $\frac{1}{i+2}$ d) $\frac{-1}{i-2}$

4. If A, B and C are invertible matrices of some order, then which one of the following is not true?

   a) $adj(AB) = (adj\ A)(adj\ B)$ b) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

   c) $\det A^{-1} = (\det A)^{-1}$ d) $adj\ A = |A|\ A^{-1}$

5. If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is

   a) 4 b) 3 c) 1 d) 2

6. If A is a non-singular matrix such that $A^{-1} = \begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$, then $(A^T)^{-1} =$

   a) $\begin{bmatrix} -1 & -3 \\ 2 & 5 \end{bmatrix}$ b) $\begin{bmatrix} -5 & 3 \\ 2 & 1 \end{bmatrix}$ c) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$ d) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$

7. If $\alpha, \beta$ and $\gamma$ are the zeros of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is

   a) $q/r$ b) $-p/r$ c) $-q/r$ d) $-q/p$

8. $\sin^{-1}\left(\frac{3}{5}\right) - \cos^{-1}\left(\frac{12}{13}\right) + \sec^{-1}\left(\frac{5}{3}\right) - \operatorname{cosec}^{-1}\left(\frac{13}{12}\right)$ is equal to

   a) 0 b) $2\pi$ c) $\tan^{-1}\left(\frac{12}{65}\right)$ d) $\pi$

9. $\sin(\sin^{-1}x) = x$ if

   a) $|x| \geq 1$ b) $|x| \leq \pi/2$ c) $|x| \leq 1$ d) $|x| < 1$

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10. In the case of Cramer's rule which of the following are correct?
    a) the system has unique solution b) $\Delta = 0$
    c) $\Delta \neq 0$ d) the system has infinitely many solutions

11. If $\cot^{-1}\left(\sqrt{\sin\alpha}\right)+\tan^{-1}\left(\sqrt{\sin\alpha}\right)=u$, then cos 2u is equal to
    a) 0 b) −1 c) $\tan 2x\alpha$ d) $\tan^2 x\alpha$

12. If $(1 + i)(1 + 2i)(1 + 3i) \ldots\ldots (1 + ni) = x + iy$, then $2.5.10\ldots\ldots (1 + n^2)$ is
    a) $x^2 + y^2$ b) $1 + n^2$ c) i d) 1

13. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is
    a) 1 b) $\sqrt{10}$ c) $\sqrt{11}$ d) 3

14. If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is
    a) 1 b) −1 c) 3 d) 9

15. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is
    a) 2ab b) $\sqrt{ab}$ c) $a/b$ d) ab

16. The number of tangent to the circle from inside the circle is
    a) 0 b) 2 real c) 2 imaginary d) can't be determined

17. If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a},\vec{c},\vec{b}]$ is equal to
    a) 1 b) 0 c) −1 d) 2

18. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar unit vectors such that $\vec{a}\times(\vec{b}\times\vec{c})=\frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ is
    a) $\pi$ b) $\pi/4$ c) $3\pi/4$ d) $\pi/2$

19. If the direction cosines of a line are $1/c, 1/c, 1/c$, then
    a) $c = \pm\sqrt{3}$ b) $c > 0$ c) $0 < c < 1$ d) $c = \pm 3$

20. The length of perpendicular from origin to the plane $\vec{r}.(3\hat{i}+4\hat{j}+12\hat{k})=26$ is
    a) 2 b) $\frac{26}{169}$ c) 26 d) $\frac{1}{2}$

## Part - II

**II. Answer any 7 questions. (Q.No.30 is compulsory)** 7 x 2 = 14

21. Prove that $\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$ is orthogonal.

22. Find a polynomial equation of minimum degree with rational coefficients having 2i+3 as a root.

23. Show that the polynomial $9x^9 + 2x^5 - x^4 - 7x^2 + 2 = 0$ has at least six imaginary roots.

24. Find the centre and radius of the circle $x^2 + y^2 + 6x - 4y + 4 = 0$

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25. Find the length of latus rectum of the parabola $y^2 = 4ax$

26. If $2\hat{i} - \hat{j} + 3\hat{k}$, $3\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + m\hat{j} + 4\hat{k}$ are coplanar, find the value of m.

27. Find the angle between the planes $\vec{r}.(\hat{i} + \hat{j} - 2\hat{k}) = 3$ and $2x - 2y + z = 2$

28. Find the principal value of $\tan^{-1}(\sqrt{3})$

29. Sketch the graph of $y = \sin x$ for $-\pi/2 \le x \le \pi/2$.

30. Simplify : $\sum_{n=1}^{100} i^n$

## Part - III

**III. Answer any 7 questions. (Q.No.40 is compulsory)** 7 x 3 = 21

31. With usual notations, in any triangle ABC, prove by vector method that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

32. Find the rank of $\begin{bmatrix} 4 & 3 & 1 & -2 \\ -3 & -1 & -2 & 4 \\ 6 & 7 & -1 & 2 \end{bmatrix}$

33. Solve $2x - y = 8$, $3x + 2y = -2$ by matrix inversion method.

34. If the equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root, show that it must be equal to $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$

35. Find the domain of $\cos^{-1}\left(\frac{2 + \sin x}{3}\right)$

36. State and prove triangle inequality.

37. If $|z| = 2$, show that $8 \le |z + 6 + 8i| \le 12$

38. Find the equation of the hyperbola with vertices $(0, \pm 4)$ and foci $(0, \pm 6)$

39. The maximum and minimum distances of the Earth from the Sun respectively are $152 \times 10^6$ km and $94.5 \times 10^6$ km. The Sun is at one Focus of the elliptical orbit. Find the distance from the Sun to the other focus.

40. Show that $[\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{a} + \vec{c}] = 2[\vec{a}, \vec{b}, \vec{c}]$

## Part - IV

**IV. Answer all the questions.** 7 x 5 = 35

41. a) Prove by Vector method that $\sin(\alpha + \beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta$

*(OR)*

b) A rod of length 1.2 m moves with its ends always touching the co-ordinate axis. The locus of a point P on the rod, which is 0.3 m from the end in contact with x-axis is an ellipse. Find the eccentricity.

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42. a) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$, show that $x^2 + y^2 + 3x - 3y + 2 = 0$

*(OR)*

b) Find the vector parametric, vector non-parametric and cartesian form of the equation of the plane passing through the points $(-1,2,0)$, $(2,2,-1)$ and parallel to the straight line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$

43. a) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

*(OR)*

b) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x + y + z = xyz$

44. a) If $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} - \hat{j} - 4\hat{k}$, $\vec{c} = 3\hat{j} - \hat{k}$ and $\vec{d} = 2\hat{i} + 5\hat{j} + \hat{k}$, verify that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}, \vec{b}, \vec{d}]\vec{c} - [\vec{a}, \vec{b}, \vec{c}]\vec{d}$

*(OR)*

b) Find the value of k for which the equations $kx - 2y + z = 1$, $x - 2ky + z = -2$, $x - 2y + kz = 1$ have (i) no solution (ii) unique solution (iii) infinitely many solution.

45. a) Solve the equation $z^3 + 8i = 0$ where $z \in C$

*(OR)*

b) Find the value of $\cot^{-1}(1) + \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) - \sec^{-1}\left(-\sqrt{2}\right)$.

46. a) Find all zeros of the polynomial $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$, if it is known that $1+2i$ and $\sqrt{3}$ are two of its zeros.

*(OR)*

b) Find the equation of tangent and normal to the ellipse $x^2 + 4y^2 = 32$ when $\theta = \frac{\pi}{4}$

47. a) Solve the system of linear equations by Cramer's rule.
$x - y + 2z = 2$, $2x + y + 4z = 7$, $4x - y + z = 4$

*(OR)*

b) Find centre, foci, vertices and directrix of the hyperbola.
$4x^2 - 24x - 25y^2 + 250y - 489 = 0$

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